



A Framework for Evaluating Mental Health Artificial Intelligence-based Conversational Agents

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Abstract

AI-based conversational agents (CA) have emerged as popular tools for addressing mental health concerns. While research has primarily focused on usability or outcome measures, there is a scarcity of studies examining the quality of CA-delivered conversations. This article emphasizes the importance of evaluating CA content by drawing parallels with the rigorous training and evaluation process for human therapists. The Thera-Turing Test (TTT) proposes a comprehensive evaluation model that scrutinizes CA conversations independently, reducing potential bias. The TTT recommends that judges rate user conversations with the CA without knowing that AI-CAs delivered those conversations. The components of the TTT include timing, conversation creation, specific vs. full assessment, selection criteria for conversations and the judges, and selecting the appropriate measures. The proposed TTT framework addresses diverse user needs by encompassing safety, common factors, specific factors, cultural awareness, referrals, and readiness for “practice” (or full-scale launch). Alternative approaches to CA evaluations are also explored, and potential critiques are addressed, highlighting the unique challenges of text-based therapy interactions. The Thera-Turing Test offers a promising avenue for ensuring the quality and effectiveness of mental health CAs, contributing to the delivery of high-quality mental health care at scale.

Keywords Artificial Intelligence · Chatbot · Conversational agent · Mental health · Evaluation · Conversation agent

Introduction

AI-based conversational agents (CA) for mental health have increased in popularity in recent years. Several CAs have been designed and tested for common mental disorders, including depression, anxiety, chronic pain, and attention-deficit/hyperactivity disorder (Fitzpatrick et al., 2017; Fitzsimmons-Craft et al., 2022; He et al., 2022; Jang et al., 2021; Sinha et al., 2022). There are CAs available in multiple languages, including English (Fitzpatrick et al., 2017), Spanish (Klos et al., 2021), and Chinese (He et al., 2022). While most of the research focuses on pilot testing or outcomes, few studies focus on the therapeutic process,

and none assess the quality of the conversations delivered by the CAs.

Evaluating the quality of content delivered by CAs is a critical aspect of development. In the field of clinical psychology, student-therapists are required to undergo several hours (e.g., 1500 hrs in California) of supervised training and receive approval from their supervisors to obtain licensure (California Board of Psychology, 2022). While some CAs have been subject to the FDA review process for treatment outcomes (e.g., Wysa for chronic pain and Woebot for postpartum depression), most CAs are not subject to a review process of their conversations. Most CA developers assess the quality of their CA conversations rather than independent reviewers, which may bias the evaluation. This raises the question of how CA content can be objectively tested. This article aims to discuss the importance of why (and how) CAs should be assessed and propose a framework for CA assessment, focusing on the therapeutic process (as a complement to the outcome research) and measuring the quality of content by independent therapists.

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Different Types of CAs and Potential Problems

Most CAs use AI to different degrees and can be categorized into three primary types, each with unique characteristics and potential issues. (1) Rule-based CAs operate on predefined rules and scripts. Rule-based CAs process text responses from the users, typically using Natural Language Processing, and guide the user through conversations that have been previously designed. Rule-based CAs usually have a limited scope and may be perceived as inadequate or annoying when responding to more complex user needs. (2) Generative Artificial Intelligence (GAI) CAs typically rely on large language models and can generate responses about a broad spectrum of topics. GAI CAs are more flexible and can adjust better to the users (than rule-based CAs). According to Lawrence et al. (2024), LLMs offer great promise by enabling flexible and natural interactions, which could help alleviate gaps in access to mental health care. However, the lack of control over what a GAI CA might say can lead to unpredictable, inconsistent, and even risky therapeutic interactions. Finally, (3) Conversational Agents (CA) with agentic architecture, guardrail and knowledge base (a repository of information on a specific topic that is used to train the bot to limit the topics it can talk about). CAs with agentic architecture seek to balance flexibility and control. However, they can present challenges, such as sudden and abrupt transitions from GAI-driven responses to scripted ones, potentially causing surprises or confusion for the user. These distinct types of CAs and their associated challenges highlight the need for thorough evaluation and understanding of how they function in therapeutic settings. By identifying and addressing these potential problems, developers and therapists can work toward creating more effective and user-friendly mental health interventions.

Importance of Evaluating Mental Health CAs

The growing popularity of mental health CAs demands a stringent evaluation. These interactions must be carefully assessed if CAs are to hold conversations with thousands or even millions of users. Relying primarily on outcomes research to assess symptom reduction is not enough. Evaluating the therapeutic process that leads to symptom reduction is equally important. To our knowledge, only a few studies focused on therapeutic processes; while they have shown promising findings in terms of creating bonds (Dosovitsky & Bunge, 2021) or therapeutic alliance (Beatty et al., 2022; Darcy et al., 2021; He et al., 2022), the mechanism by which they achieve these effects is unknown. Analyzing the content of the conversations

will allow understanding if the CA responds well to the user's specific needs, if the conversations flow naturally, and if the content of the conversations includes the treatment components that have been empirically supported in traditional studies. A better understanding of the therapeutic process delivered by the CAs can help prevent adverse outcomes and improve the efficacy of the interventions delivered through CAs. Given the potential global influence of these CA-led interventions, assessing CA-delivered conversations for mental health conditions must be rigorous and comprehensive.

In addition to analyzing the therapeutic process, some studies have provided a phased approach to integrating LLMs into mental health care, highlighting the need for careful progression from assistive to autonomous systems. Stade et al. (2024) emphasize that LLM-based CAs must evolve through stages, starting with human-supervised roles before they become fully autonomous. De Freitas et al. (2024) similarly discuss the need for monitoring AI-powered CAs, noting that even well-intentioned systems can lead to harmful outcomes, particularly if they fail to respond appropriately in crisis situations. This study builds on this by expanding the focus beyond staged deployment and risk management to a more comprehensive evaluation of CAs.

Comprehensive Assessment of Mental Health CAs

A comprehensive assessment of mental health (MH) CAs can contribute to developing and refining these tools by identifying the quality of several therapeutic skills needed for a successful treatment process. A few examples include the following elements: (1) Safety: Is the CA providing appropriate messages? Are there any harmful messages? (2) Clarity of the communications: Are the messages written clearly? (3) Rapport building: Can the CAs create a positive bond with the client? (4) Therapeutic relationship skills: Is the CA empathic, Socratic, directive, condescending, distant, or inappropriate? (5) Treatment fidelity: Is the bot incorporating and delivering essential therapeutic components when appropriate? Interestingly, most current CAs for anxiety do not include exposure activities, a critical treatment ingredient. (6) The cultural appropriateness of the conversations with the bot. However, users from different cultures send different types of messages to CAs (Chin et al., 2023). Most current CAs are designed for a universal user and do not take into consideration the cultural background of each user.

By identifying these problems, targeted solutions can be proposed, aiding in the refinement and optimization of CAs. Even a CA that yields positive outcomes can be improved. A continual quality enhancement process is necessary for delivering appropriate care at a large scale. The insights gained through a rigorous assessment of MH CAs can lead to more effective, empathic, and culturally sensitive

interactions, aligning more closely with users' therapeutic needs and expectations.

The Thera-Turing Test

This paper proposes a framework for evaluating mental health CAs beyond the traditional outcome studies, utilizing the Thera-Turing Test, a modified form of the Turing test.

The original Turing test states that if a computer can mimic human responses well enough in a conversation to the extent that an average interrogator cannot reliably distinguish between the computer and a human, then the computer can be considered to exhibit intelligence equivalent to that of a human (Turing, 1950). In contrast, the Thera-Turing Test (TTT) represents a more broad assessment. The TTT assesses if a CA can mimic therapists' behavior, provides an in-depth analysis of the CA's quality, and can be used to compare a CA with a therapist.

In this proposed evaluation model, independent therapists, separate from the CA developers, are tasked to assess the quality of CA-user conversations. However, as therapists may hold negative biases toward CAs due to fears about job security or a preference for traditional therapy methods (Sweeney et al., 2021; Xu & Zhuang, 2022), it may be recommended that therapist reviewers are deceived to mitigate the risk of biased assessments (see Fig. 1). The TTT has been developed through countless discussions between the authors and the lab members researching CAs as a result of the need to find a safe way to assess the quality of bots.

Different approaches can be used to conduct the TTT: (1) assessing the CA transcripts alone, (2) comparing the CA transcripts to those of human therapists, (3) comparing responses in each conversational turn, and (4) comparing the scores of the TTT to clinical outcomes (see Fig. 1). First, assessing the CA transcripts alone, therapist reviewers evaluate text-based therapy sessions conducted by the CAs. This can include simply reviewing the conversation using a specific measure or including the Delphi process to collect expert input on the CA conversations. The therapists' assignment involves reviewing these conversations and scoring them based on criteria such as the quality of content, conversational flow, and resemblance to traditional therapeutic dialogues. This methodology aims to create an unbiased assessment, thereby providing a more authentic and unprejudiced evaluation of the quality and therapeutic efficacy of CA-led mental health interventions. To do so, therapists are deceived into believing they are reviewing text-based therapy sessions conducted by human therapists rather than CAs. If CAs become a more regular and accepted resource in future years, the deception stage may not be needed.

Second, comparing transcripts of CA and human therapists entails providing judges with transcripts of similar topic conversations conducted by the CA with users and

contrasting them with transcripts of human therapists delivering therapy through text messages with their clients. This approach enables a direct comparison of ratings between bot-led and human-led interactions, offering insights into the relative quality and therapeutic efficacy of CA interactions. This method can directly compare the CAs competency level to that of a licensed clinician.

Third, comparing the responses of a CA and a human therapist during each conversational turn. In this scenario, both CA and human therapists will be assigned the same goal (e.g., deliver a parent training program). The user will be asked to generate an input. The CAs' response and human therapist will reply to that input, and responses can then be compared. Then, the user will generate a new input, and the process will repeat. If the CA generates responses similar to the ones of a human therapist, this could serve as evidence of the quality of the CA. In this case, the human therapist must be a licensed therapist. A slight variation of this method is to pose direct Turing test questions to judges. In this scenario, conversations would be presented, and judges would be tasked with determining whether these conversations are delivered by a human therapist or a CA. Utilizing a dichotomous question format may limit the assessment's ability to capture nuances in CA quality. However, an advantage of this approach is that if judges identify the conversation as bot-generated, they can pinpoint sections that appear robotic, facilitating potential improvements by developers.

Fourth, comparing the TTT scores to clinical outcomes can provide a better understanding of the effectiveness of the CAs. By examining clients with similar severity levels and correlating their TTT scores with their clinical outcomes, researchers can gain insights into how the CAs' performance translates into real-world therapeutic effectiveness. The expectation is that CAs with higher TTT scores will yield better clinical and process outcomes. Thus, examining the TTT with the clinical outcomes could add validity to the overall assessment of CAs. It provides a more comprehensive understanding of how the CAs' conversational abilities align with their ability to facilitate positive therapeutic changes in clients, thus informing decisions regarding their integration into clinical practice.

Adjustments to the TTT may depend on the various stages of development and deployment, such that early stages may use introductory criteria and later stages use advanced criteria—Table 1 describes the components for the TTT by stage of development and rigor criteria; below is a description of each component.

Implementation and Timing of the Thera-Turing Test (TTT)

Implementing the Thera-Turing Test (TTT) in evaluating mental health CAs should be considered a continuous

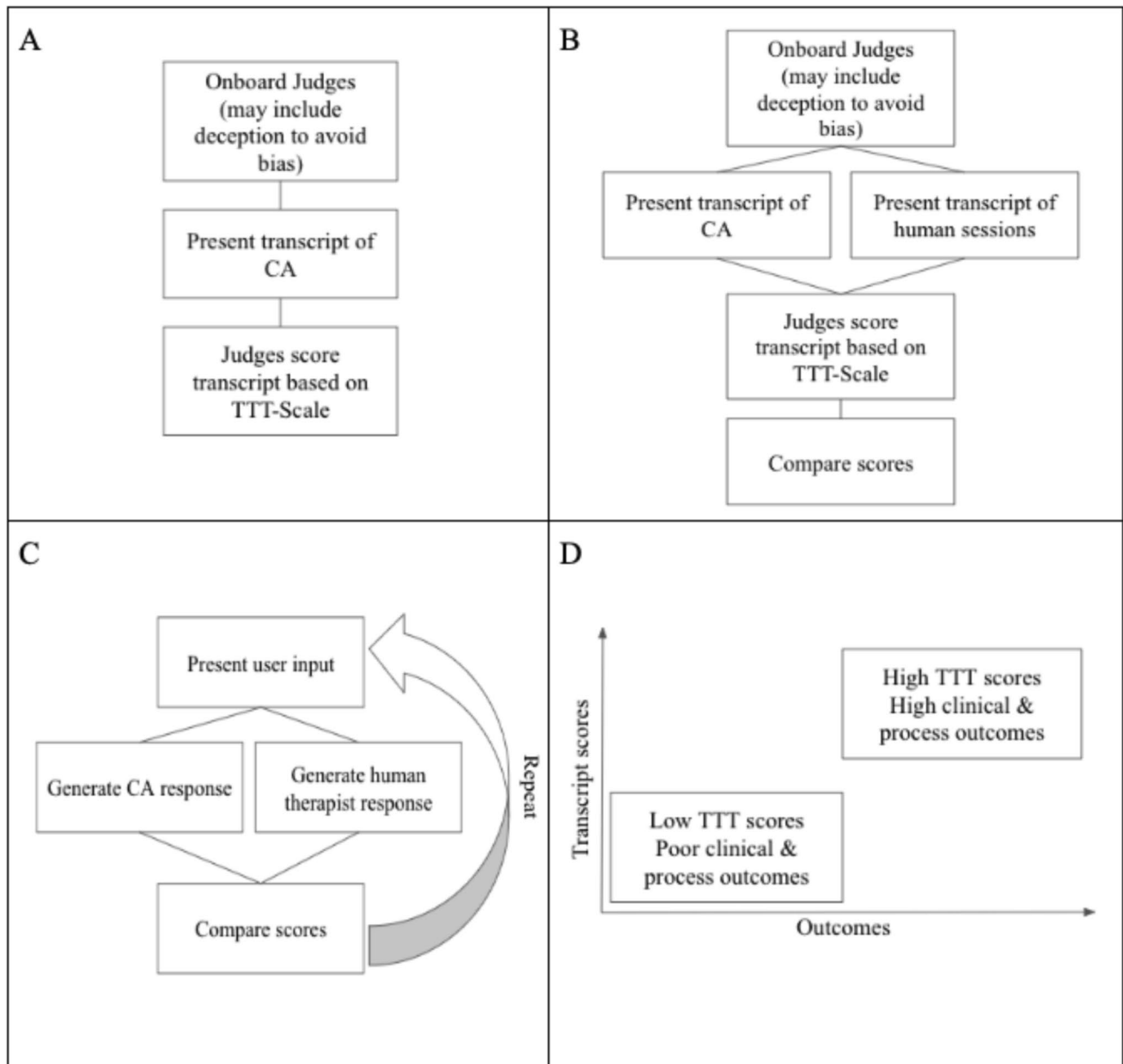


Fig. 1 TTT approaches. The figure shows the different TTT approaches that can be used to assess the CA: **A** is assessing the CA transcripts alone, **B** is comparing the CA transcripts to those of

human therapists, **C** is comparing responses in each conversational turn, and **D** is comparing the scores of the TTT to clinical outcomes

process. By strategically timing the implementation of the TTT, developers can ensure the functionality, reliability, and adaptability of mental health CAs. This approach contributes to improving and optimizing these valuable therapeutic tools, aligning them with users' evolving needs and expectations.

In early stages (e.g., beta versions), it is vital to implement the TTT during the early stages before the CA is introduced to a large audience. This proactive approach aims to prevent or reduce the risk of malfunctioning that

could affect a significant number of individuals. However, it is essential to acknowledge that there may be limitations at this stage, such as a small sample size and restricting robust statistical analysis.

In later stages (e.g., version 1.0 and further), once the CA has been deployed, the TTT should be implemented periodically. Due to the dynamic nature of technological and cultural changes, ongoing adaptations may be necessary. System crashes or societal shifts may require new modifications to the CA's behavior. Therefore, the

Table 1 Developmental stages of the TTT

TTT components	Development stage			
	Early		Later	
	Introductory criteria		Advanced criteria	
1 Implementation/timing	Implement TTT proactively		Continue TTT periodically	
2 Create conversations for the TTT	Friendly beta testers	Resistant beta testers	Regular users	Regular users
3 Specific vs. full assessment	One conversation by many users	Many conversations by a few users	One conversation/many users	Many conversations by few users
4 Selection criteria for conversations	Non-random selection		Random selection	
5 Selection of judges	Blind generalists clinicians		Blind expert clinicians in the specialized field	
6 Measurement method	Measure, treatment fidelity, common factors, and text-based specific components		Measure, treatment fidelity, common factors, and text-based specific components	
7 Outcome	Not ready	Ready for beta-testers	Ready for clinical trial	Ready for full-scale launch
	Recommend further improvements			

TTT should be repeated periodically, even after initial deployment.

Creating Conversations for the TTT

When evaluating the CA's performance, it is essential to consider the various types of users who interact with it. Various methods can be employed to create conversations that will be used in the later assessment. First, it is recommended to have “friendly or agreeable” beta testers converse with the CAs. In this case, the friendly beta testers should try to follow the flow of the conversations without crashing the bot to create the first set of conversations to begin the testing. These agreeable users can follow what might be called a “Golden path” (i.e., ideal users) by interacting with the CA in predictable ways. Second, creating conversations with “resistant” beta testers is recommended to assess how the CA would respond to more challenging clients/users. Resistant users will likely challenge the bot and deviate from the typical conversation path or script. Third, it is recommended to utilize conversations with regular users to assess how the CA responds in real-world scenarios. For the purpose of the TTT, it is vital to note that the users are aware they are interacting with a bot, and only the judges are deceived.

Specific Assessment vs. Full Assessment

The assessment of a specific conversation or the whole conversation a user has with the CA depends on what is trying to be tested, the stage of the CA development, and the type of study being conducted.

- **Specific assessment:** During the initial stages of development, it may be more appropriate to assess specific

conversations the CA has with the users. Assessing a specific conversation (e.g., psychoeducation for depression) can help refine such conversation and increase the chances of the users moving toward the following conversation. As most users tend to use digital interventions only once (Linardon & Fuller-Tyszkiewicz, 2020), it is important during the initial stages of development to ensure that the first conversations work well to increase user engagement and eventually be able to complete full treatments.

- **Full assessment:** Once specific conversations are more developed, the TTT can be conducted with all the conversations that one (or more users) have throughout treatment with the CA. Although the CA may follow a similar pathway, individual users may introduce specific details or nuances that require careful examination of how the CA responds. This approach resembles single case studies and can provide in-depth insight into how the CA adapts and interacts across multiple conversations with the same user(s).

Selection Criteria for Conversations

- **Non-random selection:** When evaluating mental health CAs, determining which conversations to test is crucial. During this developmental stage, a flexible approach to user selection can be employed that allows developers to deliberately choose specific users to participate in the testing process. This approach allows for targeted examination and may be particularly useful for identifying specific issues or features that need refinement. It is advisable to opt for a non-random selection strategy that prioritizes individuals displaying agreeable traits, thereby aligning with the overall conversational flow.

These agreeable users exhibit cooperative behavior, receptiveness, and an open-minded attitude, creating a conducive environment for assessing essential functions and features of the CA.

As the CA is more developed, the selection criteria of the conversations can gradually shift to encompass users who exhibit more resistant behaviors. This intentional shift in user selection facilitates a more rigorous examination of the CA's adaptability and responsiveness, enabling developers to identify potential areas for enhancement.

- **Random selection of conversations:** As the CA progresses to more advanced stages of development and is deployed to a larger audience, a more rigorous method involving random selection of users selection should be employed.

By randomly selecting conversations, developers can avoid potential biases in the evaluation, and it will provide a more impartial and comprehensive perspective on the CA's performance. This approach may be especially relevant for assessing CAs with a large number of users. While having more randomly selected conversations, there will be more heterogeneity in the conversations selected (some users may be friendly users, and others more resistant users). To address this methodologically will require large numbers of transcripts and judges and will result in a more robust assessment of the CA's quality.

Selection of Judges

The selection of judges should be tailored to meet the prerequisites necessary for overseeing human therapists and may vary per local regulations. Criteria for judge selection should ideally include the following: (1) licensure as a psychologist; (2) proficiency in the treatment of the specific mental health disorders addressed by the CA (e.g., depression, anxiety, and disruptive behavior); and (3) independence from any affiliation with the CA developer or company. In the evaluation process for mental health CAs, it is essential to establish specific conditions for blind experts to enable an unbiased assessment. Specifically, blind experts should be informed that (1) their task is to assess the quality of therapy delivered by a human therapist through text messages and (2) some of the text messages used by the therapist are predetermined and scripted, while others may be composed in response to specific client needs or interactions.

In the early stages of the development, generalist clinicians can judge the conversations. Ideally, in the later stages of the CA development, blind experts specialized in the topics included should assess the conversations.

Measurement Methods

Evaluating CA-driven conversations requires a structured and methodical approach that aligns with professional standards in psychology. Researchers can use already validated measures or create a specific measure depending on the aim of the CA. For example, if a CA focuses on parenting, measures of treatment fidelity to parenting programs should be implemented. However, given the specific nature of the CAs, additional aspects should be included. The appropriate measures should include a combination of text-based therapy questions and questions that supervisors use with their supervisees. More specifically, experts can adapt and utilize some of the APA competency benchmarks utilized in the training of clinical psychology doctoral students called the Practicum Competency Assessment Forms (APA, 2011). The PCAFs are used to determine if a student is ready for “pre-practicum trainee (1–2), beginning practicum trainee (3–4), intermediate practicum trainee (5–6), advanced practicum trainee (7–8), and entry-level doctoral intern (9–10).” Similarly, the TTT could be used to determine if the CA is ready to be used with beta testers (1–4), to be used in clinical studies (5–8), and ultimately to be launched at a large scale (9–10). Specific guidelines on how to rate each stage of the CA should be created and monitored by a group of experts. For example, students in a clinical psychology doctoral program are graded from 3 to 4 in beginning practicum training and are expected to demonstrate entry-level practicum competencies with substantial supervision; a grade from 5 to 6 in intermediate practicum training by demonstrating intermediate practicum competencies with routine supervision required; a grade from 7 to 8 in advanced practicum training by demonstrating advanced practicum competencies with occasional supervision needed for common situations; and a grade from 9 to 10 in entry-level to doctoral internship by demonstrating advanced competencies with increasingly autonomous clinical decision making with supervision rarely needed for common situations. A similar grading scale could be implemented to determine the CA's readiness for beta testers, clinical trials, and a full-scale launch.

Potential sections to be included in the assessment of the CA are (1) text-based specific factors, (2) common factors, (3) treatment fidelity, (4) cultural awareness, (5) safety, and (6) referrals. Text-based specific factors include (a) conversation flows naturally and (b) messages are written in a clear and concise manner. Within common factors, the following criteria could be included: (a) able to build therapeutic rapport with the client, (b) addresses the client's questions and concerns, (c) validates the client's emotions and experiences, and (d) manages client's resistance. For treatment fidelity, the criteria to consider are as follows: (a) it utilizes evidence-based practices to treat and support the user, (b) it implements each step of the treatment protocol, and (c) it

provides appropriate interventions for the user's needs. For cultural awareness, it is important to consider the following: (a) diverse cultural backgrounds when providing support and (b) tailor treatment to the client's identity when appropriate. When considering the client's safety, it is important to include criteria such as the following: (a) messages do not include content that is harmful to the client, (b) they address limits of confidentiality, and (c) they take appropriate steps to alert safety risks. Referral considerations would include (a) providing helpful referrals to the client. Once all criteria have been assessed and reviewed, the next step could be to consider the CA's readiness for practice or if further revision and evaluations are necessary.

Furthermore, the weight of each category and scale should be considered and used as needed when assessing the CA. For instance, items on quality, clarity of understanding, and safety may be more pertinent at the initial stages of the CA's development. In contrast, the evidenced-based practices and social context may need more consideration at later stages.

Outcomes of the TTT

In the final step of the evaluation process, the data serves as the foundation for determining the CA's readiness for various deployment stages: not ready, ready for beta-tester, or ready for full-scale launch. Additionally, areas for improvement can be identified, and recommendations can be made to enhance the CA's functionalities and responses. At this stage, the outcomes can help determine whether to conduct similar rounds of TTT or progress to the subsequent stage of development with more advanced assessment criteria. Even in the advanced stages of the CA, the expectation is that the TTT will be repeated periodically as part of ongoing quality assurance and ensure that the CA aligns with the highest professional standards.

Alternative Approaches to the TTT

There are a few alternative approaches to the TTT that warrant consideration, each presenting its unique advantages and disadvantages.

Licensure Examination

Evaluate CAs based on their ability to pass a licensure examination. While this proposal may seem appealing, it is essential to recognize that passing an examination does not necessarily equate to effective client treatment. An examination typically assesses knowledge and the ability to answer specific questions on various topics. In contrast, effective therapeutic dialogue involves dynamic interactions aimed at teaching clients skills and assisting them

in coping with their unique situations. Therefore, while a CA may excel in answering specific questions, its capacity to engage in nuanced and empathetic client-centered conversations remains a distinct challenge.

Data Analytics Integration

Incorporating data analytics represents a key approach to evaluating CA performance. This method involves analyzing user data to understand whether users complete conversations, find them useful, and assess overall user engagement. While data analytics offer valuable insights into user behavior, it may not provide a comprehensive view of CA quality. This parallels the therapeutic setting, where clients may not always discern the effectiveness of their treatment. Just as a supervisor assesses the quality of a therapist's work, a TTT approach with trained judges is necessary to evaluate the therapeutic quality of CA interactions. Users may have positive conversations with CAs, but these interactions may not necessarily be therapeutic in nature.

Given the complexity of CA design, not all CA are created equal, and their evaluations must reflect their diverse purposes, such as psychoeducation, emotional support, or structured interventions. This requires degrees of freedom in assessment, meaning evaluations must be flexible to fit each CA's unique goals and stage of development. Early-stage CAs may be assessed on user engagement and rapport-building, while advanced bots should be evaluated on treatment fidelity and symptomatic improvement.

This is the first attempt to conceptualize how to assess CAs, and several iterations of the framework will be needed as the field evolves. We offer the TTT as a set of flexible guidelines rather than a rigid, standardized test. Given the degrees of freedom inherent in CA design, assessment methods must be adjusted to fit the specific aims, functions, and developmental stages of individual bots. Specific formats will be required to reflect the design and purpose of each CA. Similar to research in therapy, there will not be a one-size-fits-all evaluation approach. Future implementations of the TTT will need to specify each CA's developmental stage, therapeutic goals, and intended outcomes to ensure the findings are interpreted accurately.

While these alternative approaches contribute to a comprehensive examination of CA capabilities and limitations, there still needs to be a debate with experts in the field to establish guidelines for an effective evaluation of CAs for mental health. Researchers and developers can gain a deeper understanding of CA performance by exploring various evaluation methods, paving the way for continuous improvements in mental health CA technologies.

A Critique of the Thera-Turing Test

The utility of the TTT can be questioned from several viewpoints. The primary critique of the TTT is conceptual. Conversations held by CAs are not necessarily designed to replicate human therapists. If conversations led by CAs perform less effectively than their human counterparts, they may still provide valuable assistance to numerous individuals.

Furthermore, it is possible that in the future, AI-based CAs will conduct conversations qualitatively different from the ones that human therapists have, and by using computational methods, they may enhance the therapeutic skills of evidence-based interventions (Stade et al., 2024) and eventually perform better than human therapists.

Thus, a CA does not need to be held to the same standards as a human therapist. It has been proposed that mental health professionals should focus on what generative AI can do rather than if they mimic or not human activities (Schueler & Morris, 2023). While this represents a valid critique, the authors still consider that the TTT will offer valuable information regarding the quality of the bot that should be used as a point of reference and not as the ultimate arbiter.

CAs do not need to emulate traditional therapeutic conversations due to their distinct content delivery format. Text-based therapy (conducted by humans or CAs) differs from oral communication, being asynchronous instead of synchronous, and is often delivered through multiple brief interactions across the week rather than a single weekly 50-min session. Additionally, asynchronous conversations can be interrupted, altering the urgency or context of discussions.

Several adjustments need to be implemented when translating from face-to-face sessions to text-based therapy. For example, text messages in therapy are typically shorter than live sessions, potentially limiting the depth and nuance of communication provided by both therapist and client. The lack of nonverbal communication requires other resources to communicate emotions, for example, emoticons, stickers, and other visual elements in text messages. However, these may only partially capture the full range of emotions, potentially affecting the therapeutic process. Thus, text-based interactions may need help to convey complex therapeutic techniques or interventions that rely on sensory experiences beyond text.

Additionally, the field of digital therapeutics is growing rapidly, leading to the development of new communication methods enabled by AI, including voice-to-voice interactions and video avatars. This introduces another consideration that has not yet been included in our current evaluation criteria but should be incorporated in the future. Recent studies on digital personas (Heissler et al., 2024), for example, highlight the need to extend the Thera-Turing Test (TTT) to accommodate new dimensions, such as voice tone, facial expressions, and non-verbal cues.

These critiques underscore the unique challenges involved in assessing and implementing text-based therapy. They should be thoughtfully considered in the development and evaluation of therapeutic CAs.

Conclusion

The Thera-Turing Test, as proposed, represents a pioneering framework for the assessment of CAs in the field of mental health. This method aims to objectively evaluate quality, therapeutic efficacy, and alignment with established psychological principles. Collaboration is vital in this regard. CA developers, working in conjunction with accreditation bodies such as the American Psychological Association, should formulate guidelines that accurately assess the competencies of these digital tools. Attention should be given to the unique characteristics and constraints of text-based therapeutic interactions, allowing for a nuanced understanding of how these platforms function.

Future research could further refine this approach, considering the inherent limitations and specificities of CAs. Additionally, future research should explore its application across diverse populations, ensuring that mental health CAs are effective and culturally responsive. Longitudinal studies are also needed to assess the long-term impact of CA-driven interventions on mental health outcomes. By addressing these areas, the field can enhance the reliability and cultural adaptability of AI-driven mental health tools.

The practical implications of the TTT extend to both CA developers and mental health practitioners. For developers, the TTT offers a structured framework for refining AI-driven therapeutic tools, ensuring they meet ethical and professional standards before large-scale deployment. For practitioners, the framework provides insight into the strengths and limitations of CAs, allowing for more informed integration of these tools into clinical practice. By embracing this novel assessment method, the field can guide the future development of mental health CAs, ensuring that interventions led by artificial intelligence not only mimic but also meet the stringent standards expected of human therapists. The potential benefits of this evaluation process are profound, providing a pathway for the continuous improvement of CAs, increasing their effectiveness, empathy, and cultural relevance, and ultimately contributing to the large-scale delivery of quality mental health care.

Declaration of Generative AI in Scientific Writing

The authors used Chat GPT 3.5 to write this manuscript. The authors created the idea and the initial outline of the manuscript. Chat GPT 3.5 composed the first draft; the authors edited, adding sections and iterating with Chat GPT 3.5

several times to add and improve new sections. The authors included citations when appropriate and take full responsibility for this manuscript's originality, content, and validity. All the iterations the authors had with Chat-GPT 3.5 are available upon request. Chat GPT 3.5 resulted in a thought partner during this process.

Author Contribution All authors contributed to the article's ideas and initial outline. ChatGPT was used to write the initial draft; both authors edited, adding sections and iterating with Chat GPT 3.5 several times to add and improve new sections.

Data Availability We did not analyze or generate any datasets because our work proceeds with a theoretical approach. Therefore, no data was available, no ethical approval was required, and informed consent was not needed as no participants were recruited.

Declarations

Ethics Approval This is an opinion paper, so no ethics approval is required.

Consent to Participate/Consent for Publication Informed consent was not needed for this article since no participants were used to complete this paper.

Competing interests The authors declare no competing interests.

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